

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-1987

Mitchell Dam 138 kV
December 18, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-1987. The Interconnection Customer proposes a 180.9 MW solar photovoltaic facility interconnecting at the Mitchell Dam 138 kV substation in the Virginia area. The request was received on January 18, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 703 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,828,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Mitchell Dam Substation 345/138kV TX	Haggin – Mitchell Dam 345 kV	108.2	465	64.3
Declo – New London 138 kV	Craven County Wood Energy Lp – Lone Tree 138 kV	106.0	520	5.2
S Meadow – Monroe Ne 345 kV	Channel Energy Ctr – Monroe Ne 500 kV	104.0	1,923	5.1
Morrell – Old Church 138 kV	Skaggs Island Jct 2 – Chancellor Row 345 kV	103.8	344	8.0
Northern Neck 345/138 kV Autotransformer	Chancellor Row – Skaggs Island Jct 2 345 kV	103.7	794	10.6
Shryland – Tombstone 345 kV	Tobstone – Bixby Hill 500 kV	103.1	3,552	6.7
Elmont – Muscoda Industrial 138 kV	Chancellor Row Substation – Bear Island 345 kV	102.1	513	5.7
Tippecanoe – Old Church 138 kV	Skaggs Island Junction 2 – Chancello Row 345 kV Line	102.0	385	8.0
Spring St – Wing Lane 138 kV	Skaggs Island Junction 2 – Mitchell Dam 345 kV	102.0	497	11.3
Nauvoo – Birchwood 345 kV	Fredericksburg – Dunn Siding 345 kV	102.0	3,294	18.3
Sharyland – Bixby Hill 345 kV	Tombstone – Bixby Hill 500 kV	101.6	3,496	6.8
Caver – Big Hnaford 138 kV Line	Plaza – Big Hanaford 138 kV	101.5	608	6.2
Dunn Siding – Birchwood 345 kV	Leavitt – Birchwood 345 kV	101.4	1,896	8.4
Skaggs Island Junction 2 – Tippecanoe 138 kV L	Skaggs Island Junction 2 – Chancellor Row Substation 345 kV Line	100.8	417	8.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Leavitt Substation – Birchwood Substation 345 kV line	Bea Island Substation – Dunn Siding Substation 345 kV	100.7	4,157	26.7
Kimberley – Elont 138 kV	Chancellor Row – Bear Island 345 kV	100.7	566	5.7
Skaggs Island Jct 2 – Chancellor Row 345 kV	Cancellor Row – Leavitt 345 kV	100.3	1,558	32.5
Kimberley – Shack Mills Sta 138 kV	Bear Island – Chancellor Row 345 kV line	100.0	594	5.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mitchell Dam Substation 345/138kV TX	Replace transformer with higher-rated unit	\$14,447,000	15% (\$2,111,000)
Declo – New London 138 kV	Reconductor line	\$18,676,000	1% (\$178,000)
S Meaow – Monroe Ne 345 kV	Reconductor line	\$5,305,000	2% (\$91,000)
Morrll – Old Church 138 kV Line	Reconductor line	\$9,277,000	4% (\$385,000)
Northern Neck 345/138 kV Autotransformer	Replace transformer with higher-rated unit	\$15,592,000	6% (\$882,000)
Tombstone – Sharyland 345 kV	Reconductor line	\$19,096,000	1% (\$144,000)
Elmont – Muscoda Industrial 138 kV	Reconductor line	\$1,696,000	4% (\$61,000)
Tippecanoe – Old Church 138 kV	Reconductor line	\$2,853,000	4% (\$118,000)
Spring St – Wing Lane 138 kV	Reconductor line	\$670,000	4% (\$23,000)
Nauvoo – Birchwood 345 kV	Reconductor line	\$26,790,000	1% (\$330,000)
Sharyland – Bixby Hill 345 kV	Reconductor line	\$16,777,000	1% (\$124,000)
Caver – Big Hnaford 138 kV Line	Reconductor line	\$640,000	2% (\$12,000)
Dunn Siding – Birchwood 345 kV	Reconductor line	\$16,680,000	1% (\$215,000)
Skaggs Island Junction 2 – Tippecanoe 138 kV L	Reconductor line	\$1,172,000	4% (\$49,000)
Leavitt – Birchwood 345 kV	Reconductor line	\$63,038,000	1% (\$796,000)
Kimberley – Elont 138 kV	Reconductor line	\$1,151,000	4% (\$41,000)
Skaggs Island Jct 2 – Chancellor Row 345 kV	Reconductor line	\$9,445,000	3% (\$241,000)
Kimberley – Shack Mills Station 138 kV Line	Reconductor line	\$740,000	4% (\$27,000)
	Total allocated to project		\$5,828,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.